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Version 2

🌐 Lonza 4D Nucleofection with Effective Homemade Buffer (Here using K562 with Cas9 Plasmid) even changed the name V.2

📄 Manage mention

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We use this protocol and it's working

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Abstract

Testing update system:

This protocol uses the Lonza 4D nucleofection system. While the 16 well strip kits are expensive, this allows a single strip purchase to be used for hundreds of nucleofection experiments by cleaning the strip and using homemade buffers.

The homemade buffer also works for many primary cell types as well as nucleofecting Cas9 protein/RNA (RNPs).

This is extremely efficient in K562 cells with over 99% nucleofected with a 9kb plasmid.

The homemade buffer used here was originally used and published in:
<https://cancerres.aacrjournals.org/content/71/23/7291>. It was called Amaxa V.

Guidelines

This is one buffer solution to try nucleofecting cells in. Also OptiMEM often works well.

The Porteus Lab also has a homemade buffer that they have published:
K562 electroporation solutions.

- Solution I: Dissolve 2g ATP•Disodium Salt and 1.2g $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ in 10mL nuclease-free UltraPure water, filter sterilize, make 20 μL aliquots, and store at -20°C for up to two months.
- Solution II: Dissolve 6g KH_2PO_4 , 0.6g NaHCO_3 , and 0.2g glucose in 500mL nuclease-free UltraPure water, adjust pH to 7.4, filter-sterilize, make 1mL aliquots, and store at 4°C for up to 3 months.
- Mix 20 μL solution I with 1mL solution II (see Reagent Setup). This makes for 10 electroporations (100 μL each)



Materials

MATERIALS

☒ Sodium phosphate monobasic monohydrate **Merck MilliporeSigma (Sigma-Aldrich) Catalog #S9638**

☒ HEPES **Merck MilliporeSigma (Sigma-Aldrich) Catalog #H6147**

☒ Sodium succinate, dibasic, hexahydrate **Bio Basic Inc. Catalog #SB0889.SIZE.500g**

☒ Magnesium chloride hexahydrate **Merck MilliporeSigma (Sigma-Aldrich) Catalog #M2670**

☒ Gibco Penicillin-Streptomycin (10,000 U/mL) (Pen/Strep) **Fisher Scientific Catalog # 15-140-122**

☒ FBS **Invitrogen - Thermo Fisher**

☒ Potassium chloride **Merck MilliporeSigma (Sigma-Aldrich) Catalog #P9333**

☒ Disodium phosphate **Merck MilliporeSigma (Sigma-Aldrich) Catalog #S7907**

☒ IMDM **Gibco - Thermo Fisher Scientific Catalog #12440053**

☒ 4D-Nucleofector System with X Unit **Lonza Catalog #AAF-1002X**

☒ 4D-Nucleofector® X Kit S (32 RCT) specific for cell type **Lonza Catalog #V4XC-1032**



1 Make Nucleofection Buffer (LonzaBuf1):

test

Make stock solutions. Stocks can be stored RT for a long time (we use ours longer than 6 months)

Combine stock solution and vacume filter through 0.45um filter. This should be stored at 4C. Discard after 3 months or when you see white precipitate forming.

Stock Solution	Preperation of Stock Solution	Vol of stock to add to 15ml falcon tube				
H2O		6 ml				
Sodium phosphate buffer pH 7.2, 0.5 M	Mix 7 parts of 1 M NaH ₂ PO ₄ with 18 parts of 1 M Na ₂ HPO ₄ and 25 parts of water. Check pH. If pH needs to be adjusted, use the two phosphate stock solutions and not HCl or NaOH. (7ml + 18ml + 25ml = 50 ml)	1.8 ml				
KCl 1 M		50 µl				
MgCl ₂ 1 M		100 µl				
HEPES 0.2 M	Just dissolve in water, no pH adjustment necessary.	1 ml				
Sodium succinate 0.24 M		1 ml				
filter the solution		Total volume ~10 ml				

2 Prepare your Nuclofection Condtions:

Each nucleofection conditions requires ~2 ug of plasmid (Cas9 plasmid is about 9kb). Smaller plasmid nucleofect easier. If the plasmid has a fluorescence marker or some other signal, do a titration experiment to see your ideal amount (0.2- 4ug)

Add the plasmid for each condition to a well in the pcr strip. Ideally the volume is less than 5ul for reactions in the Londza 4D 16-well strips but I have added up to 10ul of

plasmid solution successfully (final reaction condition would then be 20ul of cells + 10ul plasmid solution). Plasmids should be in TE or water.

3 **Count and pellet K562 Cells**

cells should be in log phase of growth

Use 500k K562 cells/ reaction well. (1 million also works fine)

Take the total cells needed for all reactions. Pellet at 300 rcf for 4 min.

4 **Resuspend in LonzaBuf1:**

Remove supernatant from cells.

Add 20ul LonzaBuf1 for each 500k cells and resuspend.

i.e. for 6 reactions take 3 million cells, resuspend in 120ul LonzaBuf1 (lonzabuf1 should be room temp).

5 **Nucleofect**

Add 20ul LonzaBuf1/Cells onto the side of each of the pcr tubes. Do not put it in the bottom or your pipette tip will be contaminated between the different conditions. After adding cells to the side of the pcr tube. Tap the pcr tube on the hood bench until cells fall into the bottom.

Using multi-channel pipette. Pipette cell/plasmid mix up and down 2-4x. Transfer the cells/dna to the nucleofection strip. Tap strip so that there is no airbubbles at the bottom.

Nucleofect with FF-120

This should be done relatively quickly to keep the cells from being in the electroporation medium any extra time than necessary.

6 **Move Cells into Well**

As quickly as possible after nucleofection add 70-80ul IMDM/10%FBS/1%PenStrep to each well of the strip (cells/dna/buffer that was just nucleofected). Then transfer the entire 100ul solution to a well of a 12-well rack containing 1ml of IMDM/FBS/PenStrep media.

7 **Wash Strip**

Take strip, put it in 50ml falcon tube. Fill falcon and strip with 70% ethanol. Leave for 5 min. Then put strip on bench in hood to dry. Can also leave it in 70% ethanol and dry directly before using.

Strips can then be used 40x or more times.

